

An enzyme kinetics assay reveals that $V_{\max} = 100 \text{ nmol/s}$ and $K_M = 10 \text{ }\mu\text{M}$. For this assay, what substrate concentration is required for a reaction rate V_0 of 75 nmol/s ?

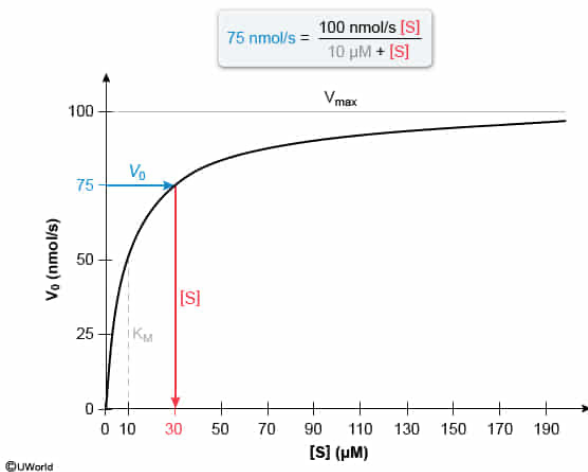
- ☐ A. $20 \text{ }\mu\text{M}$ [15%]
- ✓ ☒ B. $30 \text{ }\mu\text{M}$ [50%]
- ☐ C. $50 \text{ }\mu\text{M}$ [12%]
- ☐ D. $75 \text{ }\mu\text{M}$ [20%]

Correct

51% answered correctly

Explanation:

Michaelis-Menten equation shows the relation between $[S]$ and V_0



The **Michaelis-Menten equation** describes the **velocity V_0** of any enzyme-catalyzed reaction as a **function of substrate concentration $[S]$** . The general form of the equation is

$$V_0 = \frac{V_{\max}[S]}{K_M + [S]}$$

where **$V_{\max}$ is the maximum velocity** and **K_M is the $[S]$ at which $V_0 = \frac{1}{2}V_{\max}$** . When any **three of the four values** (V_0 , $V_{\max}$, K_M , and $[S]$) are **known**, the **remaining value can be determined** mathematically.

The question states that $V_{\max} = 100 \text{ nmol/s}$, $K_M = 10 \text{ }\mu\text{M}$, and $V_0 = 75 \text{ nmol/s}$.

The easiest way to solve this problem is to substitute each of the given answer choices into the Michaelis-Menten equation along with the given parameters

$V_{\max}$ and K_M and determine which substrate concentration yields the desired reaction rate V_0 . **When $[S] = 30 \mu\text{M}$ (Choice B), the equation yields**

$$V_0 = \frac{100 \frac{\text{nmol}}{\text{s}} \times 30 \mu\text{M}}{10 \mu\text{M} + 30 \mu\text{M}}$$
$$V_0 = 100 \frac{\text{nmol}}{\text{s}} \times \frac{30 \mu\text{M}}{10 \mu\text{M} + 30 \mu\text{M}}$$
$$V_0 = 100 \frac{\text{nmol}}{\text{s}} \times \frac{30 \mu\text{M}}{40 \mu\text{M}}$$
$$V_0 = 100 \frac{\text{nmol}}{\text{s}} \times \frac{3}{4}$$
$$V_0 = 75 \frac{\text{nmol}}{\text{s}}$$

Note that the $V_{\max}$ term is a reaction *rate* whereas the K_M and $[S]$ terms are *concentrations*. Therefore, the units of $V_{\max}$ (nmol/s) are not converted to match the units of K_M or $[S]$ (μM) because they describe different aspects of the reaction.

As an alternative, the given values of $V_{\max}$, K_M , and V_0 can be substituted into the equation to **solve for $[S]$ directly**. Both methods show that for this assay, a **substrate concentration of $30 \mu\text{M}$ yields a reaction rate of 75 nmol/s** .

(Choice A) Substituting into the Michaelis-Menten equation shows that if $[S]$ were **$20 \mu\text{M}$** , V_0 would be 66.7 nmol/s .

(Choice C) Substituting into the Michaelis-Menten equation shows that if $[S]$ were **$50 \mu\text{M}$** , V_0 would be 83.3 nmol/s .

(Choice D) Substituting into the Michaelis-Menten equation shows that if $[S]$ were **$75 \mu\text{M}$** , V_0 would be 88.2 nmol/s .

Educational objective:

The Michaelis-Menten equation $V_0 = \frac{V_{\max}[S]}{K_M + [S]}$ relates the observed rate V_0 of an enzyme-catalyzed reaction to the substrate concentration $[S]$ by the constants $V_{\max}$ and K_M . If any three of the values V_0 , $V_{\max}$, $[S]$, and K_M are known, the

fourth can be determined by solving the equation.

Time spent:0 sec

Subject:
Biochemistry

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System:
Enzymes